Agnel Charities' Fr. C. Rodrigues Institute of Technology Department of Information Technology Practical No. 7 Implementation of views	
Name of Expt.	Implementation of views

Theory

<u>View</u>

A database view is known as a "virtual table" that allows you to query the data in it. Views can be considered as virtual tables. Generally speaking, a table has a set of definition, and it physically stores the data. A view also has a set of definitions, which is build on top of table(s) or other view(s), and it does not physically store the data.

Advantages of creating view.

1. Limits data access to specific user.
2. Provide extra layer of security

Disadvantages

1. Performance decreases
2. Dependency on table.

The syntax for creating a view is as follows:

CREATE VIEW view_name AS
SELECT column_name(s)
FROM table_name
WHERE condition

Parameters:

view_name: It specifies the name of the VIEW that you want to create in MySQL.

WHERE conditions: It is also optional. It specifies the conditions that must be met for the records to be included in the VIEW.

To see the created VIEW:

Syntax: **SELECT** * **FROM** view_name;

<u>MySQL Update VIEW</u>

In MYSQL, the ALTER VIEW statement is used to modify or update the already created VIEW without dropping it.

Syntax: **ALTER VIEW** view_name **AS**
SELECT columns
FROM table
WHERE conditions;

The syntax for updating a view is as follows:

CREATE OR REPLACE VIEW view_name AS

SELECT column_name(s)
FROM table_name
WHERE condition
The syntax for dropping a view is as follows:
DROP VIEW view _name.

SQL> create table emp (empno number(6), ename varchar2(20)not null, job varchar2(10) not null, deptno number(3), sal number(7,2)); Table created.

empno	Ename	job	deptno	sal
1	Mathi	AP	1	10000
2	Arjun	ASP	2	15000
3	Gugan	ASP	3	15000
4	Karthik	Prof	4	30000
5	Akalya	AP	5	10000
6	suresh	lect	6	1 8000

Q1: The organization wants to display only the details of the employees those who are ASP.
Ans: SQL> create view empview as select * from emp where job='ASP';
View created.
SQL> select * from empview;

EMPNO	ENAME	JOB	DEPTNO	SAL
2	Arjun	ASP	2	12000
3	Gugan	ASP	2	20000

Q2: The organization wants to display only the details like empno, empname, deptno, deptname of the employees. Ans: SQL> create view empview1 as select ename,sal from emp;
View created.
Q3: Execute the DML commands on the view created.
Ans: SQL> select * from empview;
EMPNO ENAME JOB DEPTNO SAL
- - - -

CONCLUSION: Student are able to create virtual tables.

VIVA QUESTIONS

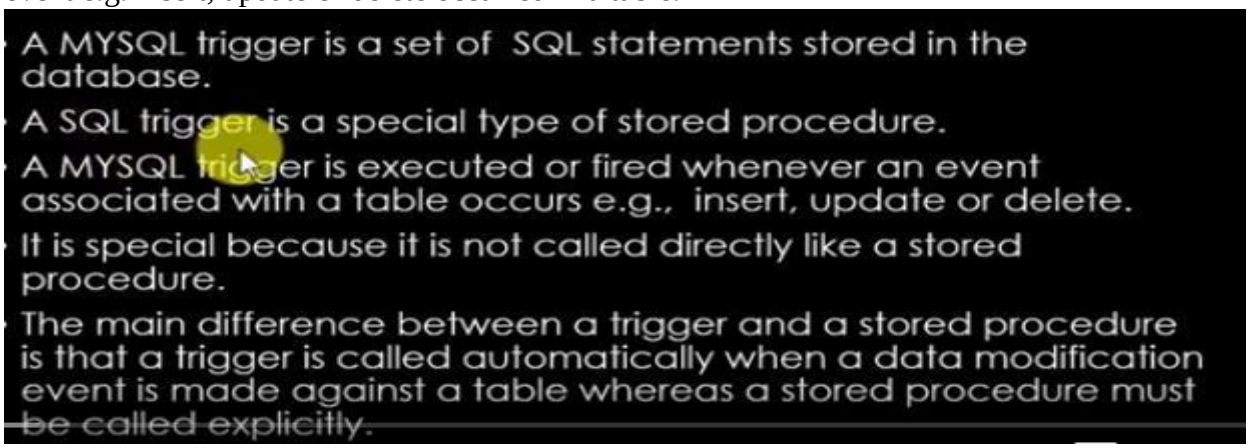
1. Define view.
2. What is the need of a view?
3. List out the advantages of views.
4. What is the syntax for creating a view?
5. How can you insert data into a view?
6. How can you update data into from a view?
7. What is the syntax for deleting a view?
8. What is the syntax for renaming the columns of a view?

Agnel Charities'
Fr. C. Rodrigues Institute Of Technology
Department of Information Technology
Practical No. 7 **Implementation of Triggers**

Name of Expt.	**Implementation of Triggers**
Aim of expt	**TRIGGER FOR DISPLAYING default value**

Theory

- A trigger or database trigger is a stored program executed automatically to respond to a specific event e.g. insert, update or delete occurred in a table.



- The database trigger is powerful tool for protecting the integrity of the data in your MySQL databases.
- In MySQL, a trigger is a set of SQL statements that is invoked automatically when a change is made to the data on the associated table. A trigger can be defined to be invoked either before or after the data is changed by INSERT, UPDATE or DELETE statement.

Define maximum six triggers for each table.

- BEFORE INSERT – activated before data is inserted into the table.
- AFTER INSERT – activated after data is inserted into the table.
- BEFORE UPDATE – activated before data in the table is updated.
- AFTER UPDATE – activated after data in the table is updated.
- BEFORE DELETE – activated before data is removed from the table.
- AFTER DELETE – activated after data is removed from the table.

Syntax

```
CREATE TRIGGER trigger_name
  AFTER/BEFORE INSERT
     ON table_name FOR EACH ROW
     BEGIN
    --variable declarations
    --trigger code
    END;
```

Parameter:

trigger_name: name of the trigger to be created.

AFTER/BEFORE INSERT: It points the trigger after or before insert query is executed.

table_name: name of the table in which a trigger is created.

Example

```
mysql> create table sample(id int primary key, age int, adult int);
Query OK, 0 rows affected (1.10 sec)
mysql> DELIMITER  //
mysql> SELECT * FROM sample;
    -> //

mysql> create trigger adult_check before  insert
    -> on sample
    -> for each row
    -> begin
    -> if new.age>18 then
    -> set new.adult=1;
    -> else
    -> set new.adult=0;
    -> end if;
    -> end//
Query OK, 0 rows affected (0.13 sec)

mysql> insert into sample (id,age) values(1,12);
    -> //
Query OK, 1 row affected (0.05 sec)

mysql> select * from sample;
    -> //
+----+-----+-------+
| id | age | adult |
+----+-----+-------+
|  1 |   12 |     0 |
+----+-----+-------+
1 row in set (0.00 sec)

mysql> insert into sample (id,age) values(2,22);
    -> //
Query OK, 1 row affected (0.08 sec)

mysql> select * from sample;
    -> //
+----+-----+-------+
| id | age | adult |
+----+-----+-------+
|  1 |   12 |     0 |
|  2 |   22 |     1 |
+----+-----+-------+
2 rows in set (0.00 sec)
```

CONCLUSION: Successfully implemented before insert trigger on sample table **.**

VIVA QUESTIONS

1. Define database triggers.
2. What are the pars of triggers and it uses?
4. What are the types of trigger?

5. Compare before trigger and after trigger.
9. What is the syntax for DROP a trigger?
10. List out the some situations to apply before and after triggers.